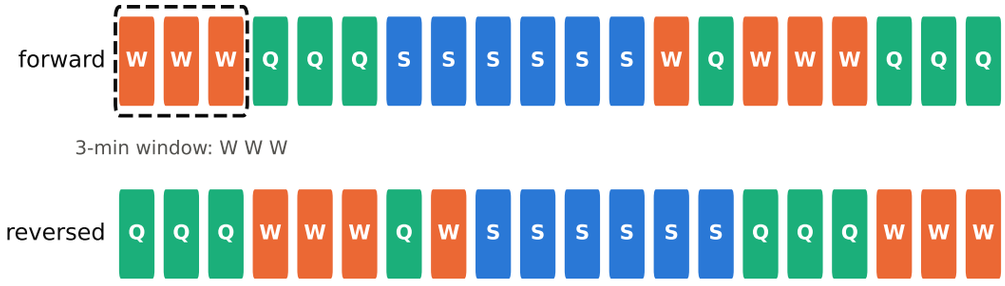
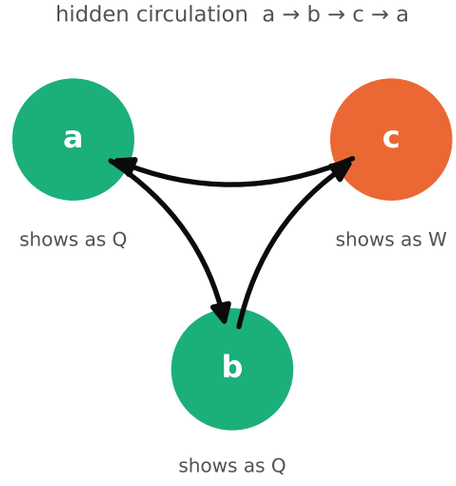
a The same record played forwards and backwards (illustration)



Pairs (2-min windows): every Q→W is eventually matched by a W→Q,
so pair counts can look balanced, as if nothing runs in a preferred direction.

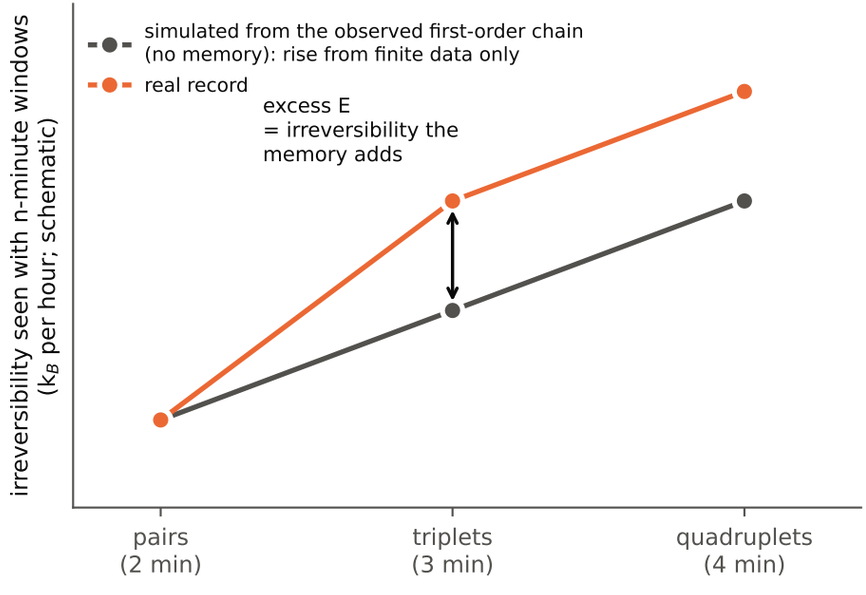
Triplets (3-min windows) also see the ORDER inside a run: e.g. "W Q Q" (slowing down) vs "Q Q W" (abrupt start). If one is more common than its reverse, the movie has an arrow of time that pairs cannot show. That extra arrow is a lower bound on dissipation.

b Why it happens: a hidden cycle seen through a lumped camera



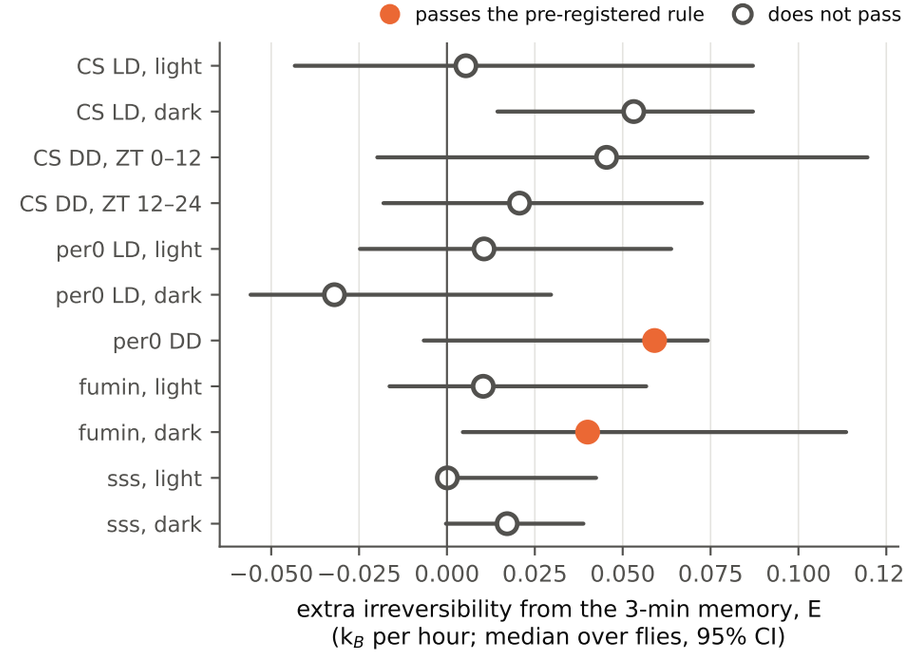
The camera cannot tell a from b.
Observed: … W Q Q W Q Q W …
- Q→W and W→Q pairs balance
- but inside each Q run the fly always goes a then b, so runs "end the same way" more than they "start" that way: triplets are asymmetric

c How it is measured (schematic)



A first-order chain gives a straight line: each extra minute adds the same amount. Raw curves rise even without memory (estimator bias), so only the gap to the matched simulation counts.

d What we found (20p, data)



fumin dark: the excess sits in windows containing sleep (5-min criterion not excluded).
per0 DD: present in wake-only windows. CS LD dark: CI excludes 0 but too few single flies (4%) pass.